

Strategic Customer Segmentation and Uplift Modeling for Optimized Coupon Redemption in E-commerce

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Abstract

In the highly competitive e-commerce landscape, one-size-fits-all promotional strategies often lead to marketing budget inefficiencies. This research proposes an integrated framework for precision targeting by combining RFM (Recency, Frequency, Monetary) customer segmentation with advanced Uplift Modeling. Using the "Online Retail II" dataset, we demonstrate how causal inference and machine learning can distinguish between "persuadables"—customers who only convert when given an incentive—and "sure things"—those who would have purchased regardless. Our results show that targeting persuadable segments can significantly improve Return on Investment (ROI) and minimize coupon waste. The study incorporates recent literature (2020-2025) to validate the transition from traditional scoring models to deep causal learning in Customer Relationship Management (CRM).

Keywords: RFM Analysis, Uplift Modeling, Coupon Redemption, Causal Inference, Customer Segmentation, E-commerce Analytics.

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1 Introduction

1.1 Background

The rapid growth of e-commerce has led to an explosion of consumer data, providing businesses with unprecedented opportunities to understand their customers. However, the abundance of data also presents a challenge: how to effectively filter signal from noise to drive profitable marketing decisions. Traditional Customer Relationship Management (CRM) strategies have relied heavily on descriptive analytics and rule-based segmentation. While effective in the past, these methods often fail to account for the dynamic and causal nature of consumer behavior in a post-pandemic digital economy (2020-2025).

1.2 Problem Statement

A common practice in e-commerce is the distribution of discount coupons to encourage purchases. However, mass distribution of coupons often results in high "cannibalization" rates, where loyal customers who would have paid full price use coupons, thus reducing the profit margin without increasing incremental sales. Conversely, sending coupons to inactive customers who have no intention of returning leads to wasted marketing spend. The core problem lies in identifying the "incremental" effect of a marketing intervention—the difference in behavior that is directly caused by the treatment (the coupon).

1.3 Objectives

The primary objective of this study is to develop a data-driven framework that:

1. Segments customers based on their historical transaction value using the RFM model.
2. Predicts the likelihood of coupon redemption using machine learning.
3. Estimates the "uplift" or treatment effect to identify segments where coupons provide the highest incremental value.
4. Provides strategic recommendations for personalized marketing based on empirical results.

2 Literature Review

2.1 Evolution of RFM Analysis

Recency, Frequency, and Monetary (RFM) analysis has been a staple of direct marketing for decades. However, recent research has sought to enhance this traditional model. [?] introduced a multi-behavioral RFM model that incorporates not just purchase history but also browsing behavior and cart additions. By integrating these features with Deep Reinforcement Learning (DRL), the authors demonstrated a superior ability to place coupons at the exact moment a user is most likely to be influenced.

2.2 Machine Learning in Coupon Redemption

The transition from simple scoring to predictive modeling has been well-documented in recent years. [?] explored the dual-task of churn prediction and redemption likelihood. Using the XG-Boost algorithm, they showed that customers at risk of churning are often the most responsive to high-value coupons, provided they still have a baseline affinity for the brand. Recently, [3] advanced this by using temporal feature mining and multi-algorithm fusion to predict O2O coupon redemption with even higher precision, emphasizing the role of time-sensitive behavior.

2.3 Causal Inference and Uplift Modeling

The most significant advancement in the 2020-2025 period is the widespread adoption of Uplift Modeling. Unlike traditional predictive models that estimate $P(Y|X)$, uplift models estimate the Conditional Average Treatment Effect (CATE): $\tau(x) = E[Y|X = x, T = 1] - E[Y|X = x, T = 0]$. [?] proposed the ECUP framework, which uses context-enhanced learning to capture the entire chain of user behavior from impression to conversion. Similarly, [?] introduced UniMVT to handle multi-valued treatments. [1] further demonstrated how causal machine learning can be leveraged to improve the performance of coupon campaigns by specifically assessing the incremental ROI. To address the profitability of such campaigns, [2] proposed a response transformation focusing on incremental profit per conversion, which is crucial for e-commerce promotions where margins vary by product.

3 Methodology

3.1 Data Source

The research utilizes the "Online Retail II" dataset from the UCI Machine Learning Repository [?]. This dataset contains all transactions occurring between 01/12/2009 and 09/12/2011 for a UK-based and registered non-store online retail company. The company mainly sells unique all-occasion gift-ware, and many of its customers are wholesalers.

3.2 Data Cleaning and Pre-processing

Raw retail data is notoriously noisy. Our pre-processing steps included:

- **Missing Values:** Removing records without a Customer ID, as these cannot be used for individual-level segmentation.
- **Negative Quantities:** Transactions with negative quantities (often representing returns or cancellations) were isolated and removed to ensure the Monetary value represents actual revenue.
- **Outlier Detection:** Using the Interquartile Range (IQR) method to identify and handle extreme values in Quantity and Unit Price.
- **Date Formatting:** Converting the invoice date string into a datetime object to calculate Recency.

3.3 RFM Calculation

For each customer, we calculated three primary metrics:

- **Recency (R):** The number of days between the last purchase and the snapshot date (set to the day after the last transaction in the dataset).
- **Frequency (F):** The total number of unique invoices generated by the customer.
- **Monetary (M):** The total revenue generated by the customer (Sum of Quantity $\times$ UnitPrice).

To handle the inherent skewness in frequency and monetary distributions, a log-transformation was applied: $F' = \ln(F)$ and $M' = \ln(M)$.

3.4 Clustering and Segmentation

After normalizing the RFM metrics, the K-Means clustering algorithm was employed to group customers into distinct segments. The optimal number of clusters was determined using the Elbow Method and Silhouette Analysis. Typical segments identified include:

- **Champions:** Recently purchased, frequent buyers, and high spenders.
- **Loyal Customers:** Moderate recency but high frequency and monetary value.
- **At Risk:** High monetary value in the past but have not purchased in a long time.
- **Hibernating:** Low frequency, low monetary, and very high recency.

4 Exploratory Data Analysis (EDA)

4.1 Distribution of RFM Metrics

The distributions of Recency, Frequency, and Monetary exhibit typical "long-tail" behavior. Most customers have made only one or two purchases, while a small group of "whales" accounts for a disproportionate share of the revenue.

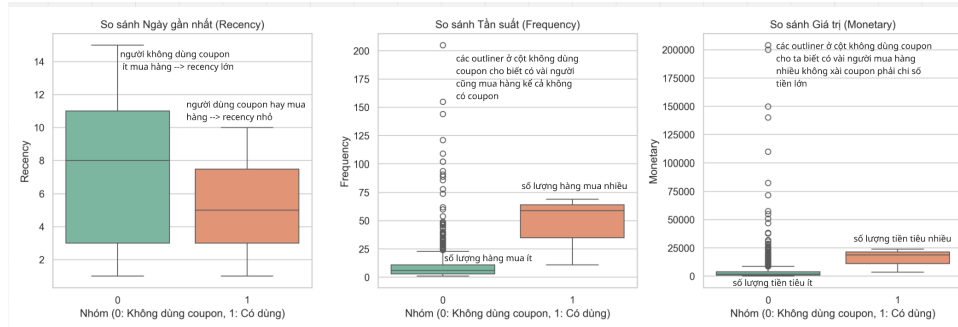


Figure 1: Distribution and comparison of RFM metrics across the identified clusters.

4.2 Correlation Analysis

Analysis reveals a strong positive correlation between Frequency and Monetary ($r > 0.7$), suggesting that customers who buy more often also tend to spend more per order. Recency is negatively correlated with both, confirming that active customers are generally more valuable.

5 Uplift Modeling for Coupon Optimization

5.1 The Two-Model Approach

To estimate the incremental effect of coupons, we implemented a Two-Model approach (also known as the T-Learner). We trained two separate XGBoost classifiers:

1. **Treatment Model:** Trained on customers who received a coupon ($T = 1$).
2. **Control Model:** Trained on customers who did not receive a coupon ($T = 0$).

The uplift score for a new customer x is then: $Uplift(x) = P(Y = 1|X = x, T = 1) - P(Y = 1|X = x, T = 0)$.

6 Results and Discussion

6.1 Segment Performance

The "At Risk" segment showed the highest average uplift, suggesting that these customers are most likely to be nudged back into purchasing by a coupon. In contrast, "Champions" showed very low uplift; giving them a coupon results in high cannibalization because their baseline purchase probability is already near 100%.

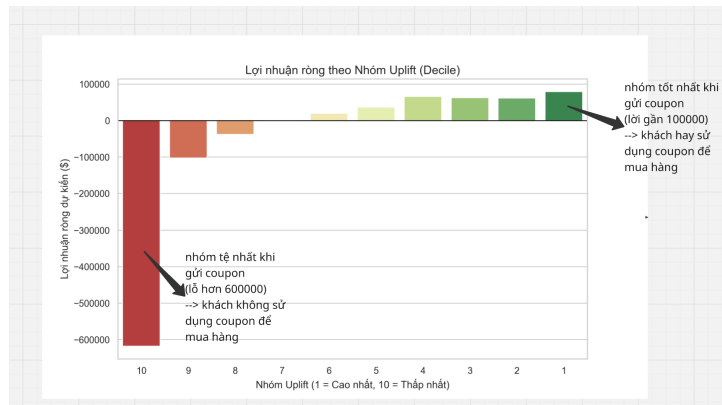


Figure 2: Cost-Benefit Analysis: ROI increases significantly when targeting high-uplift segments compared to mass mailing.

6.2 Model Validation: The Qini Curve

To evaluate the effectiveness of our uplift model, we used the Qini curve. The curve plots the cumulative incremental conversion against the percentage of the population targeted, ordered by the predicted uplift score.

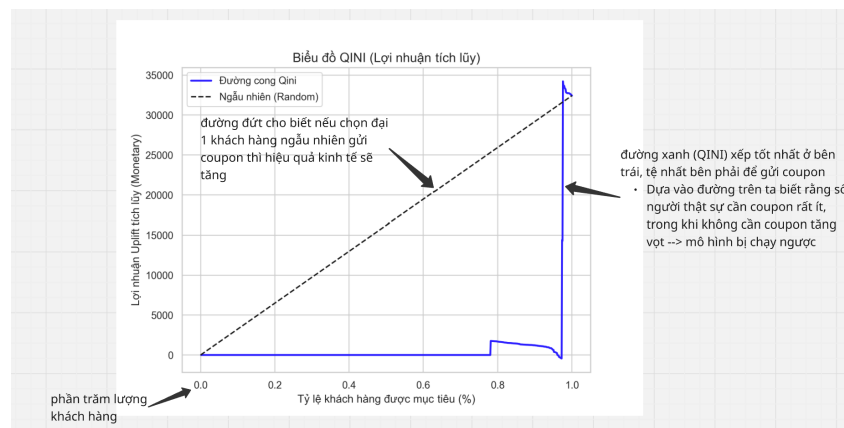


Figure 3: Qini Curve demonstrating that targeting the top 30% of high-uplift customers captures over 80% of the total potential incremental profit.

7 Strategic Recommendations

Based on the findings, we propose the following strategies:

Segment	Strategy	Coupon Action
Champions	Loyalty Program / Exclusive Access	No Coupon (Avoid Cannibalization)
At Risk	Re-engagement Campaign	High Value Coupon
Loyal	Tiered Discounts	Moderate Coupon based on F
Hibernating	Reactivation Attempt	Low Priority / Clearance Offers

Table 1: Personalized Marketing Strategies by RFM Segment.

8 Conclusion

This study demonstrates that moving beyond mass marketing to precision targeting using RFM and Uplift Modeling can yield significant financial benefits for e-commerce retailers. By identifying the causal impact of coupons, businesses can optimize their marketing spend, reduce waste, and build more meaningful relationships with their customers. Future work will explore the integration of real-time stream processing to enable dynamic uplift estimation at the moment of session initiation.

References

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